

COL703/7203

13th August

Set Representation of CNF formula

$$(P_1 \vee P_2) \wedge (P_3 \vee \neg P_4 \vee P_5) \wedge (\neg P_6 \vee P_7)$$

Set representation
for the given formula

$$\underbrace{\{P_1, P_2\}} \wedge \underbrace{\{P_3, \neg P_4, P_5\}} \wedge \underbrace{\{\neg P_6, P_7\}}$$

↑
disjunction of these
literals

$$P_1 \wedge (\neg P_2 \vee P_3) \wedge P_4$$

$$\{ \{ P_1 \}, \{ \neg P_2, P_3 \}, \{ P_4 \} \}$$

$$P_1 \wedge (\neg P_2 \vee P_3 \vee \underline{\underline{\neg P_2}}) \wedge (P_4)$$

$$\{ \{ P_1 \}, \{ \neg P_2, P_3 \}, \{ P_4 \} \}$$

An empty clause is equivalent
to false and is denoted by
 $\{\}$ or $\square$.

An empty formula is equivalent
to true.

Propositional Resolution

$$(P_1 \vee P_2 \vee \neg P_3)$$

complementary
pair of
literals

$$(P_3 \vee P_4)$$

$$(P_1 \vee P_2 \vee P_4)$$

$$\{P_1, P_2, \neg P_3\}$$

$$\{P_3, P_4\}$$

$$(C_1 \setminus \{\neg P_3\}) \cup (C_2 \setminus \{P_3\})$$

$$\{P_1, P_2, P_4\}$$

resolution

Let C_1 and C_2 be clauses. A

clause R is called a resolvent of

C_1 and C_2 if there are

complementary literals $l \in C_1$

$\bar{l} \in C_2$

such that

$$R = \underbrace{(C_1 \setminus \{l\}) \cup (C_2 \setminus \{\bar{l}\})}_{=}$$

$\bar{l}$ is the complementary literal of l
i.e., if $l = p$, $\bar{l}$ is $\neg p$
if $l = \neg p$, $\bar{l}$ is p .

$$F = \boxed{C_1, C_2, \dots, C_k}, C_{k+1}, C_{k+2}, C_{k+3}, \dots$$

The diagram shows a set of clauses F enclosed in a box. From the bottom of this box, three arrows point to the clauses C_{k+1} , C_{k+2} , and C_{k+3} . Additionally, there is an arrow pointing from the ellipsis $\dots$ to the right, indicating a continuation of the sequence.

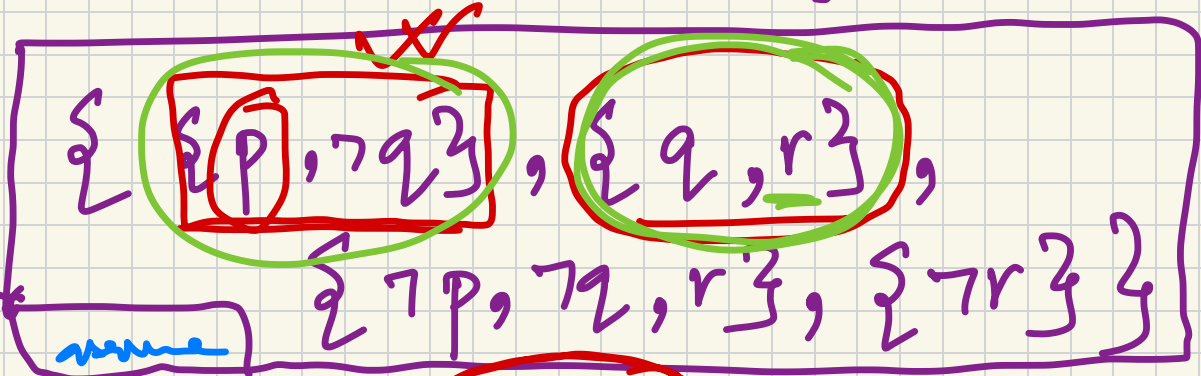
A derivation of a clause C from a set of clauses $\boxed{F}$ is a sequence

$$C_1, C_2, \dots, \underline{\underline{C_m}} = C$$

such that each C_i either belongs to F or is obtained by resolving C_j and C_k where $j, k < i$.

A derivation of the empty clause is called a refutation of F . qur

Example



1. $\{p, \neg q\}$
2. $\{q, r\}$ assum
3. $\{ \neg p, \neg q, r \}$ assum
4. $\{ \neg r \}$ assum

5. $\{p, r\}$ assum
6. $\{ \neg q, r \}$
7. $\{ r \}$
8. $\{ \}, \square$

resolution 1, 2
resolution 5, 3
resolve 6, 2
resolve 7, 4

A resolution refutation of a formula can be seen as a proof that the formula is unsatisfiable.

Let F be a set of clauses. Then

$\text{Res}(F)$ is defined as

$$\text{Res}(F) = F \cup \{R : R \text{ is a resolvent of two clauses in } F\}$$

$$\text{Res}^0(F) = F$$

$$\text{Res}^1(F) = F \cup \{R\}$$

$$\text{Res}^2(F) = \text{Res}^1(F \cup \{R\}) \dots$$

$$\text{Res}^0(F) = F$$

$$\text{Res}^{n+1}(F) = \text{Res}(\text{Res}^n(F))$$

$$\text{Res}^*(F) = \bigcup_{n \geq 0} \text{Res}^n(F)$$

(Resolution Lemma)

Claim

Let F be a CNF formula
(represented as a set of clauses).

Suppose (R) is a resolvent of two clauses
 $C_1, C_2 \in F$ then $F \equiv \underline{\underline{F \cup \{R\}}}$.

$$p \vee r \rightarrow \left[(p \vee \neg q) \wedge (q \vee r) \right] \wedge (p \vee r)$$

$$\rightarrow (p \vee \neg q) \wedge (q \vee r)$$

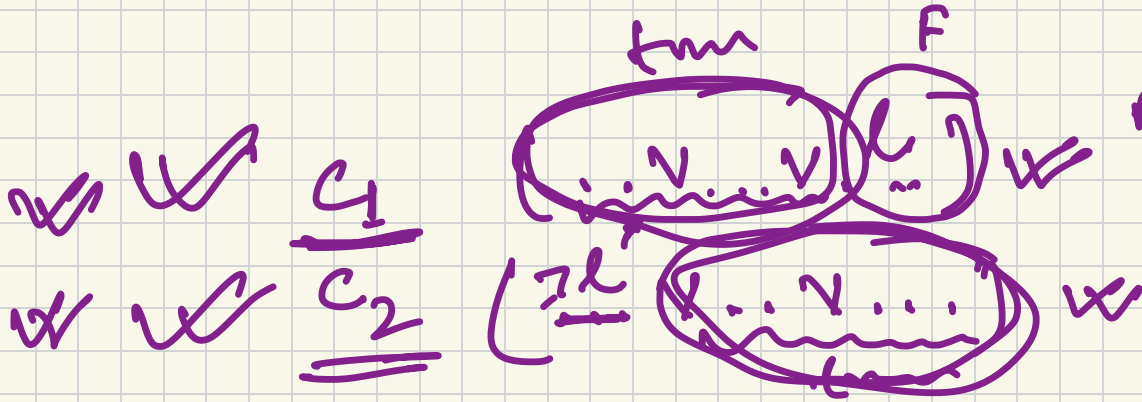
$$\textcircled{F} = \{ \{ p, \neg q \}, \{ q, r \} \}$$

$$R = \{ p, r \}$$

$$\textcircled{F \cup R} = \{ \{ p, \neg q \}, \{ q, r \}, \{ p, r \} \}$$

Any assignment A that satisfies F will also satisfy $F \cup \{R\}$

obtained by
resolving C_1, C_2
 $\in F$



A must have
given a truth value to l

✓ Soundness If there is a derivation of $\square$ from F then F is unsatisfiable.

(Refutation)

✓ Completeness

If F is unsatisfiable then there is a derivation of $\square$ from F .

